ARLADI PUZZLES

Rainbow Carnival

Brain Adventurers

AGES 9–11 • SET 05

16 fresh picture-puzzle activities



Reason • Plan • Discover

Big ideas start here.

Work step by step. Check that your answer follows every clue. Sketch or write working when it helps.



This book belongs to

_____

1 LOOK closely

Notice the pictures and the clues.

2 TRY a plan

Point, draw, count, or sketch to help your thinking.

3 CHECK and smile

A new try is part of learning. There is no timer.

For grown-ups

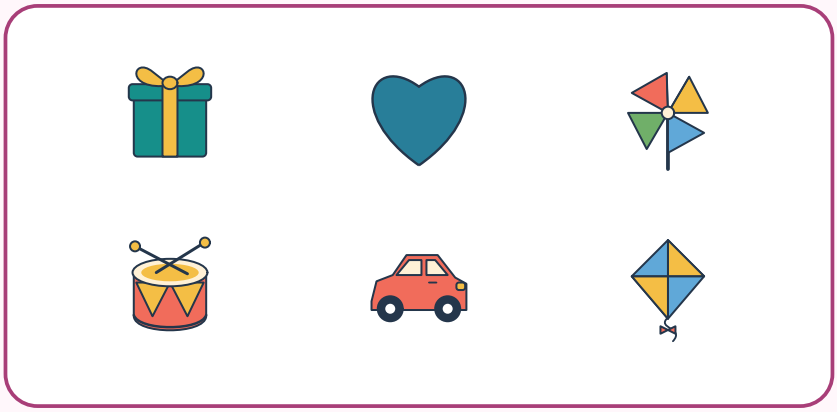
Combine visual reasoning with number patterns, spatial challenges, data, and multi-step logic.

Each activity has a matching answer card on pages 19–22. Drawing and memory activities include guidance for checking together. Let your child choose a comfortable pace.

PRINT: A4 portrait in color. For Letter paper, choose Fit.

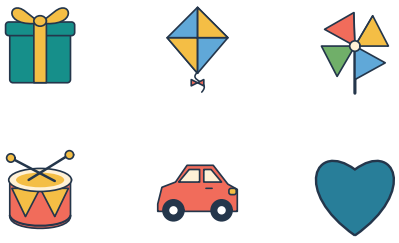
Remember six places

Study the top box. Cover it with paper, then choose the box with exactly the same six pictures in the same positions.

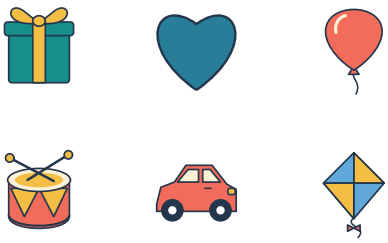


COVER THE TOP BOX

A



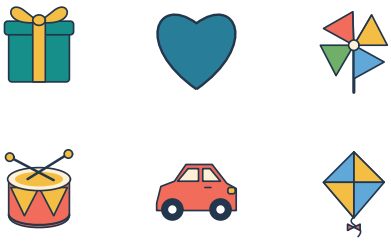
B



C



D



Read the picture votes

Each small star means 4 votes. How many more votes did kite and balloon get together than gift?

★ = 4 votes



★ ★ ★ ★ ★ ★ ★

kite _____



★ ★ ★

balloon _____



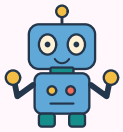
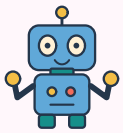
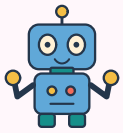
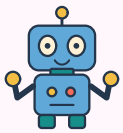
★ ★ ★ ★ ★

gift _____

How many more votes? ☐

Crack the number machine

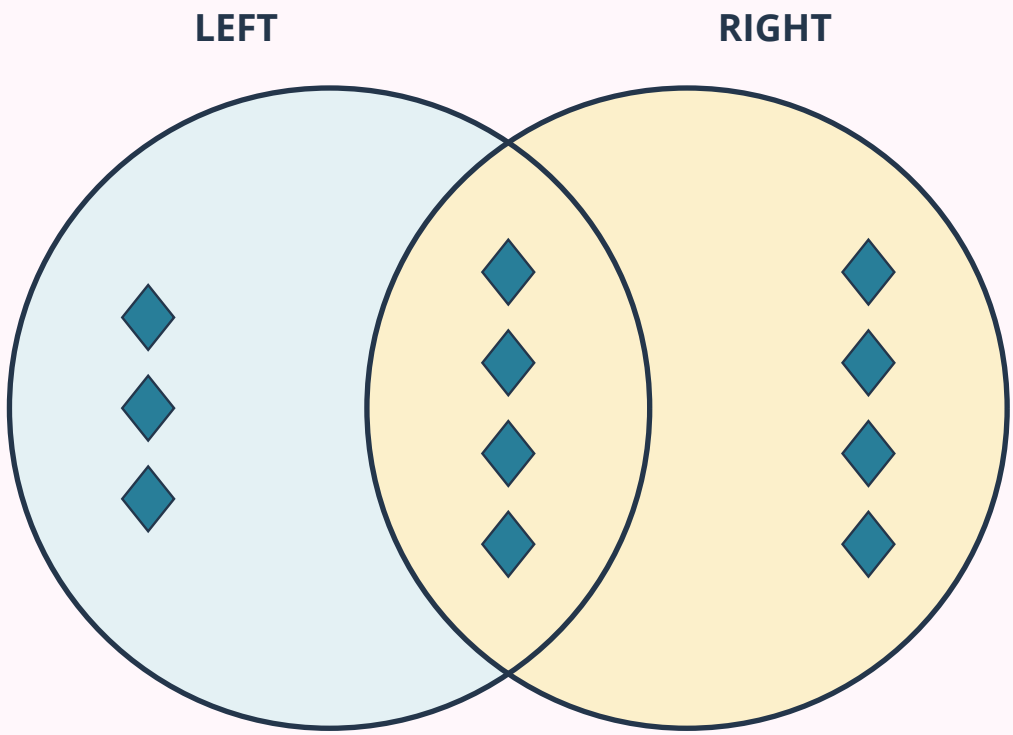
The rule is: multiply by a whole number from 2 to 5, then add a whole number from 1 to 9. The same rule fits every row.

INPUT		OUTPUT
6	→  →	27
9	→  →	36
12	→  →	45
15	→  →	

The missing output is:

Overlap point hunt

Each diamond inside exactly ONE circle earns 2 points. Each diamond inside BOTH earns 4 points. Outside diamonds earn 0. Find the total.



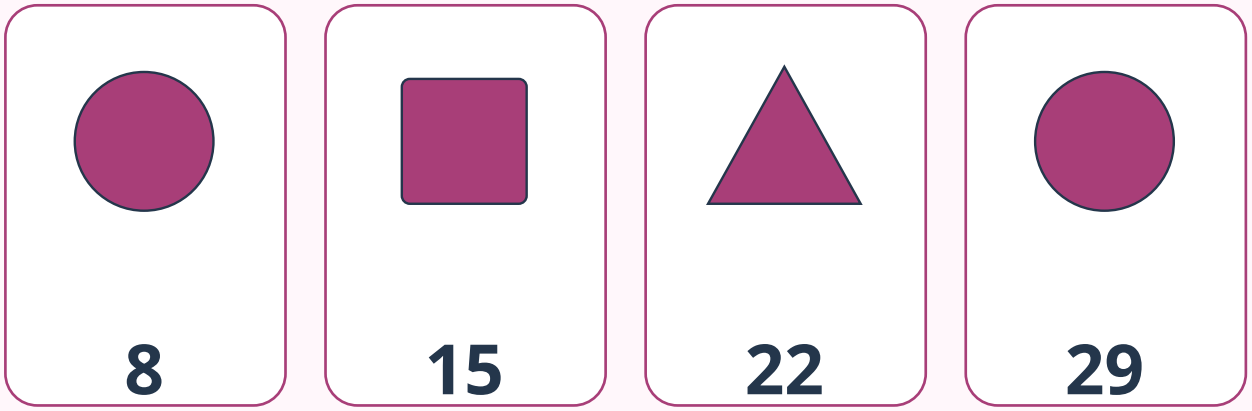


Outside

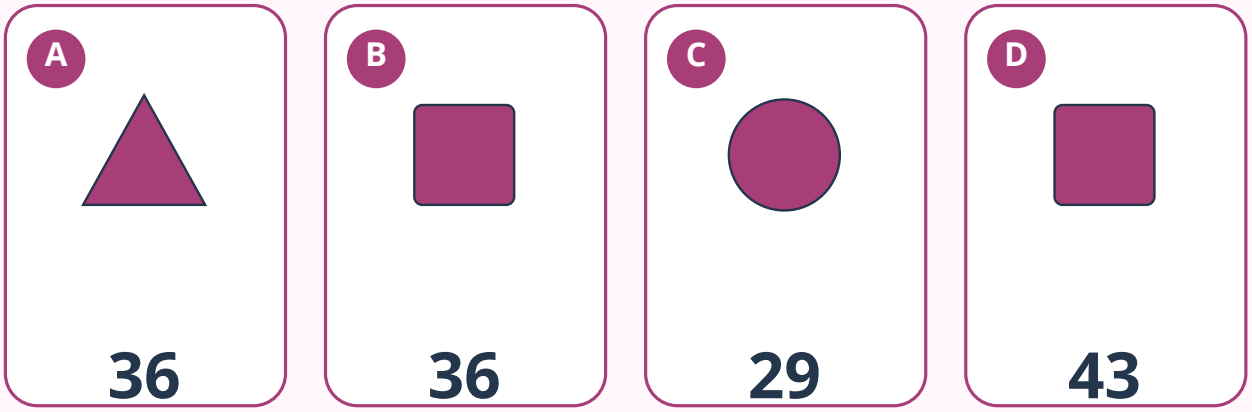
Total score:

Two rules at once

Find the next card. Follow both the shape rule and the number rule.

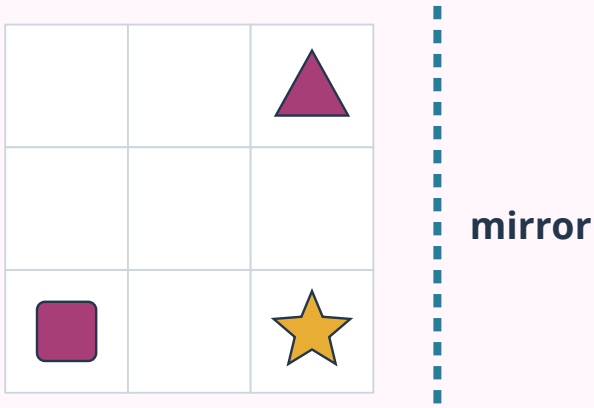


Which card comes next?

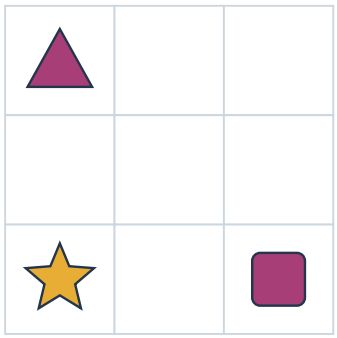


Move the whole group

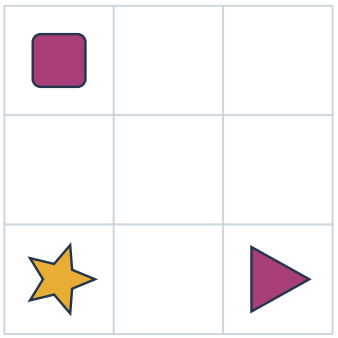
Reflect the WHOLE group in an upright mirror. Which option is the result?



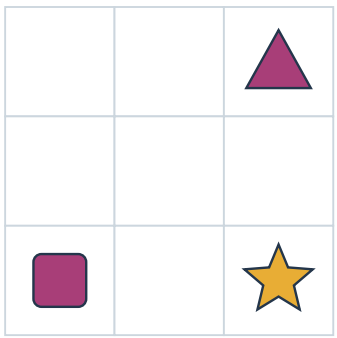
A



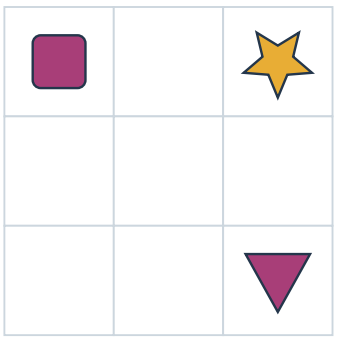
B



C



D



Picture equation quest

The same picture always has the same value. Solve the clues, then the final sum.

 +  = 14

 +  = 20

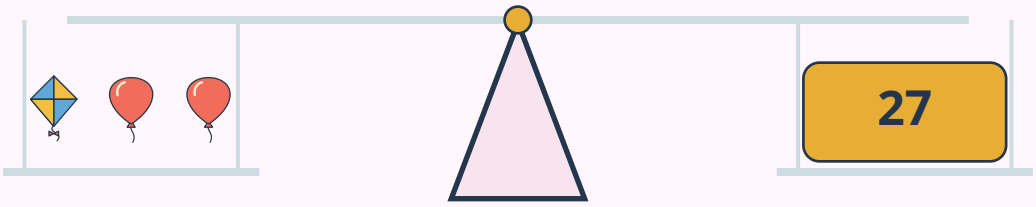
 +  = 22

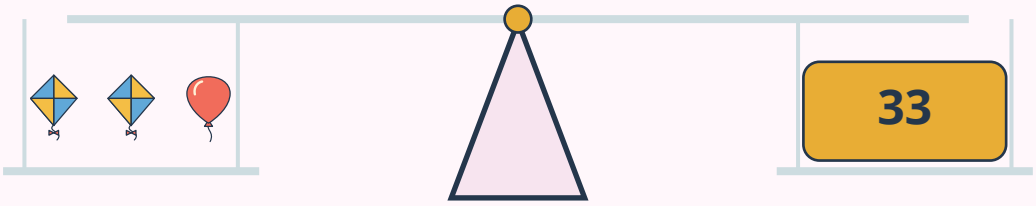
Now find:

 +  +  = ☐

Balanced treasures

Both scales balance. The yellow weights are in grams. Find the combined mass of one of each picture.





One of each:  **+**  **=** ☐ g

PUZZLE 09

Four by four Sudoku

Write 1, 2, 3 and 4 once in every row, every column and every bold 2-by-2 box.

	2	3	
4	3	2	
			3

Check all three: rows, columns, small boxes.

Four friends, four finds

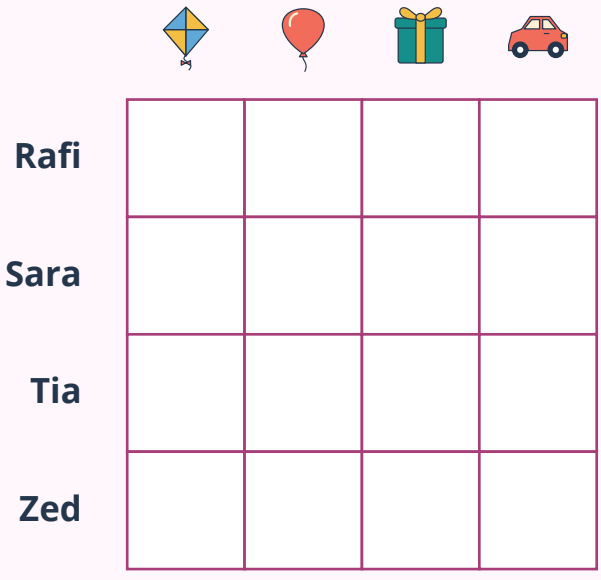
Each friend chose one different object. Use the clues and tick one square in each row and column.

Object order: left to right



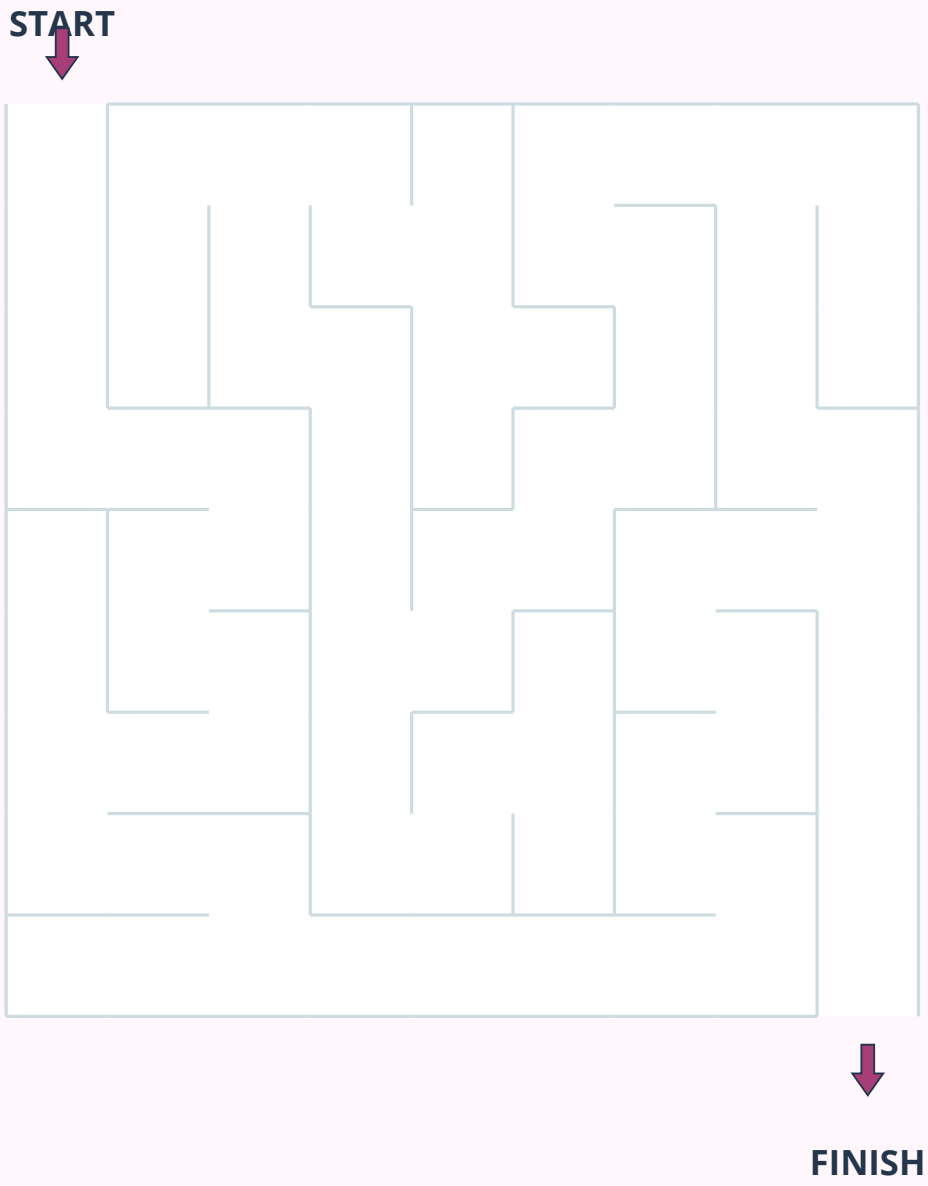
kite balloon gift car

1. Tia chose the object immediately left of Zed's.
2. Tia and Sara chose neighboring objects.
3. Rafi chose an object somewhere left of Sara's.



The winding way

Find the only route from START to FINISH. Stay in the passages and never cross a wall.



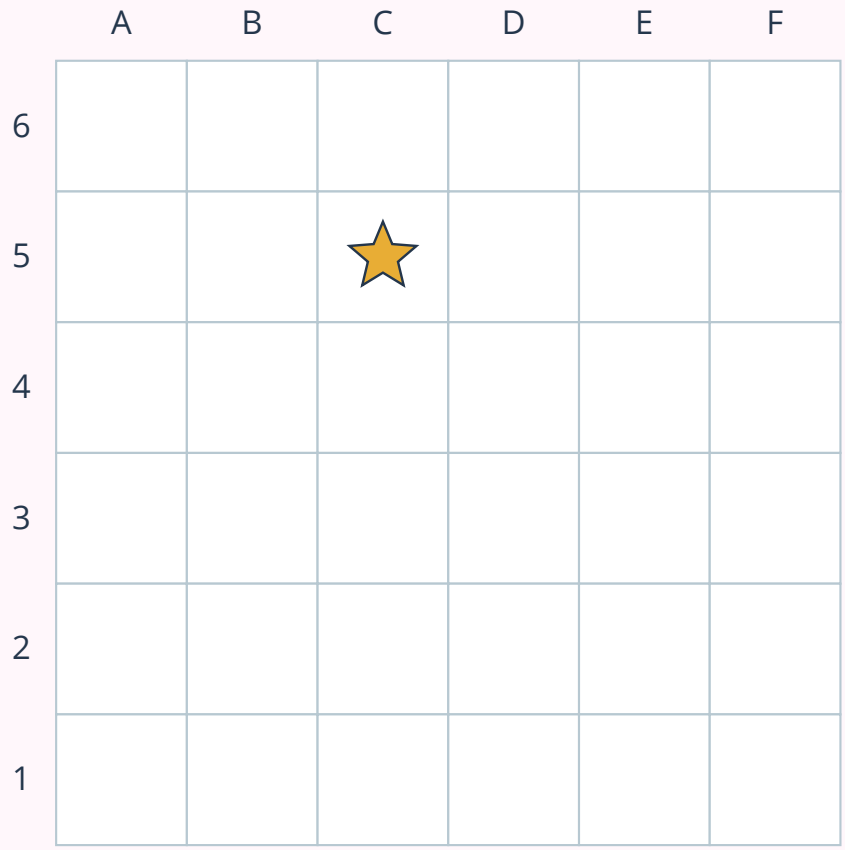
Coordinates in motion

Start at the star. Follow the arrows in order. One step moves to the next square. Name the finish coordinate.

START: C5

1 ↑ 1 | 2 ← 2 | 3 ↓ 1 | 4 → 2

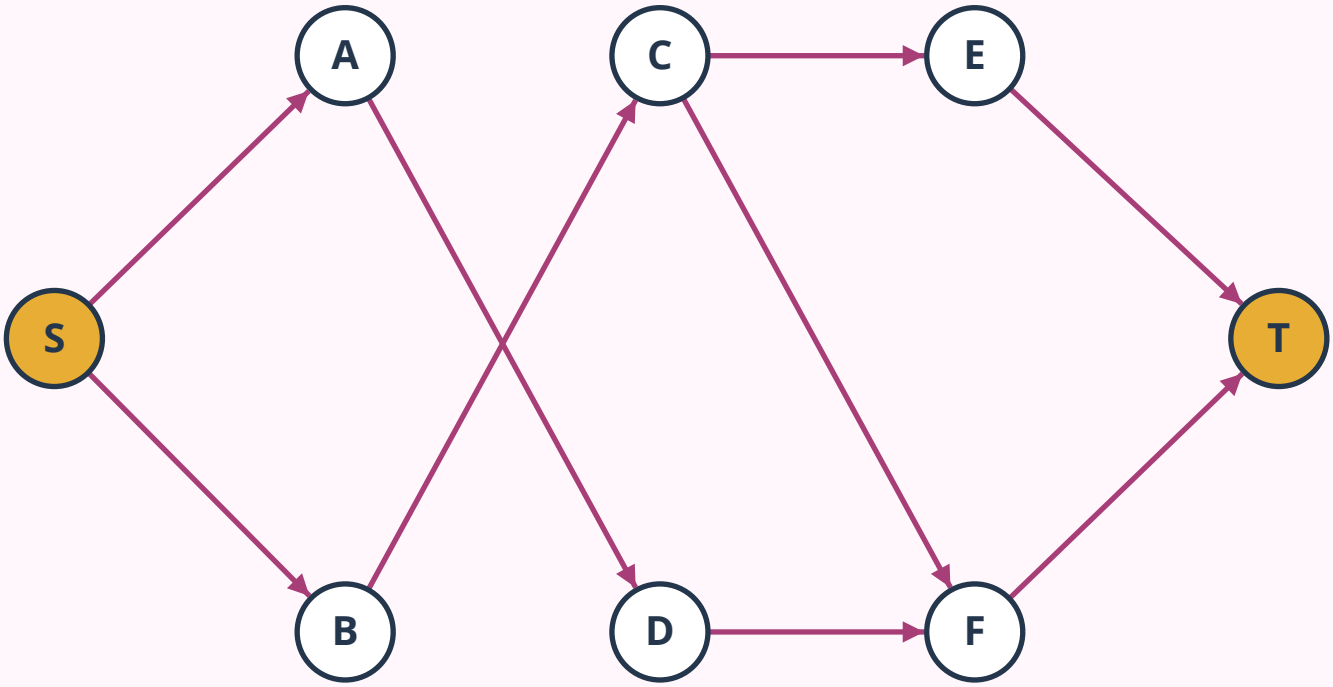
5 ↓ 3 | 6 ← 2 | 7 ↓ 1



Finish coordinate: _____

How many different paths?

Count all routes from S to T. Follow arrow directions. Change paths only at circles, even where lines cross.



Write routes here, such as S-A-D-F-T:

Total routes:

Share the mosaic

There are 28 tiles. 3/14 get dots and 6/14 get stripes. No tile gets both. How many stay blank?

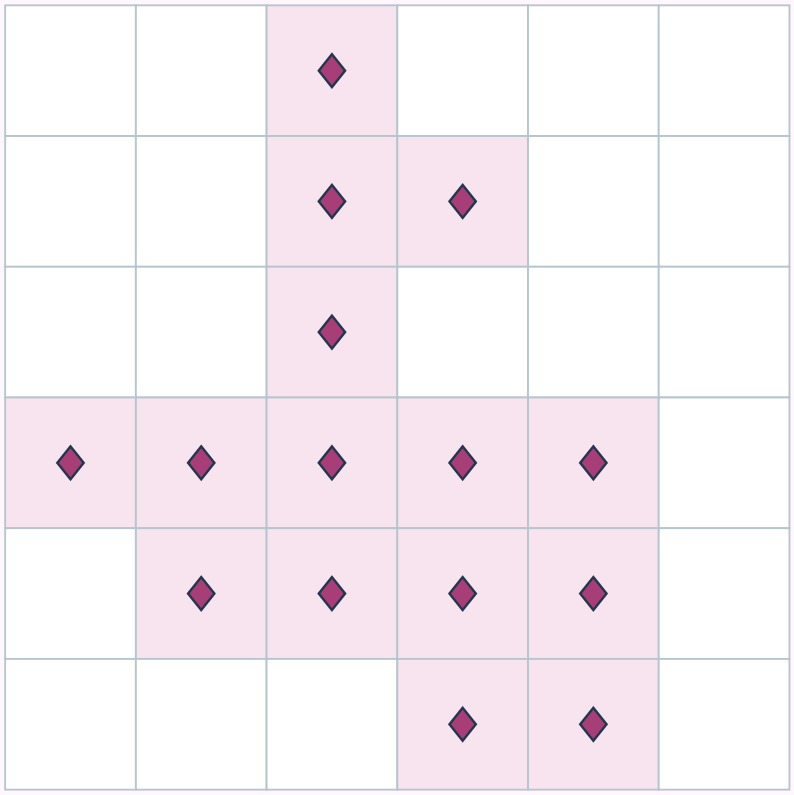
Draw dots and stripes. Keep the groups separate.

You may mark any tiles to help you count.

Blank tiles:

Walk around the edge

Each small square has side length 1 unit. Find the perimeter around the shaded shape. Shared edges inside do not count.



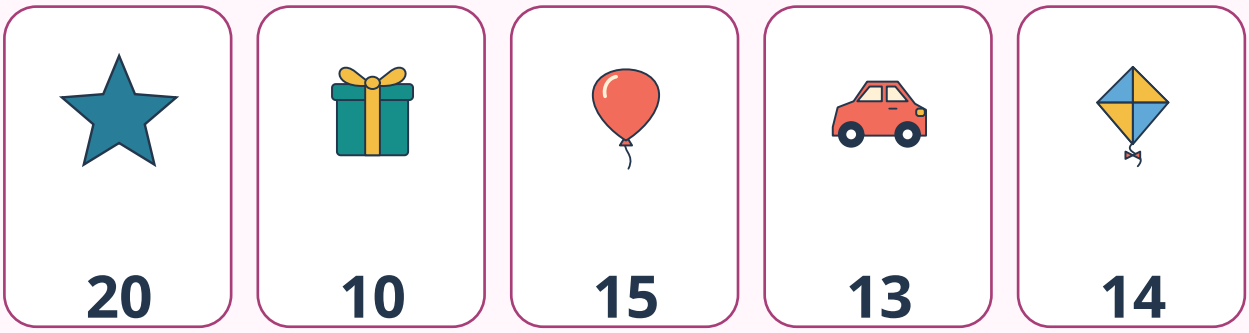
1 unit

Perimeter: _____ units

The shifted picture code

Each picture has a number. Add 5 to that number, then change it to a letter using A=1, B=2 and so on.

PICTURE KEY



Decode these pictures from left to right:



A 1	B 2	C 3	D 4	E 5	F 6	G 7	H 8	I 9	J 10	K 11	L 12	M 13
N 14	O 15	P 16	Q 17	R 18	S 19	T 20	U 21	V 22	W 23	X 24	Y 25	Z 26

Check your discoveries

Puzzles 1–4 • Use the page number to find each puzzle.

01 • PAGE 3

Remember six places

D

Top row: gift, heart, pinwheel. Bottom row: drum, car, kite.

02 • PAGE 4

Read the picture votes

20 votes

Totals are 28, 12 and 20. Calculate 28 + 12 - 20.

03 • PAGE 5

Crack the number machine

54

Multiply by 3, then add 9. So 15 x 3 + 9 = 54.

04 • PAGE 6

Overlap point hunt

30 points

7 diamonds are in exactly one circle; 4 are in both. 7 x 2 + 4 x 4 = 30. Ignore the 3 outside.

Check your discoveries

Puzzles 5–8 • Use the page number to find each puzzle.

05 • PAGE 7

Two rules at once

B: square, 36

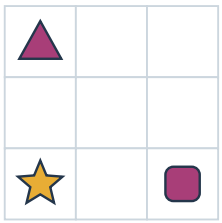
The shapes repeat in groups of three. Add 7 each time to the number.

06 • PAGE 8

Move the whole group

A

Every item moves with the group. Keep all three shapes in their correct transformed positions.



07 • PAGE 9

Picture equation quest

25

kite = 7, balloon = 13, gift = 9. Use those values in the last sum.

08 • PAGE 10

Balanced treasures

20 g

The first picture is 13 g and the second is 7 g. They satisfy both balanced scales.

Check your discoveries

Puzzles 9–12 • Use the page number to find each puzzle.

09 • PAGE 11

Four by four Sudoku

Completed grid shown.

There is exactly one completion that meets all three rules.

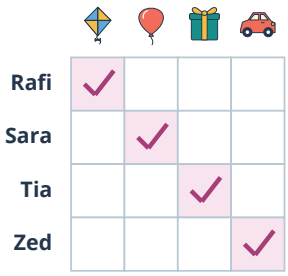
3	4	1	2
1	2	3	4
4	3	2	1
2	1	4	3

10 • PAGE 12

Four friends, four finds

Matching grid shown.

Each friend chose exactly one object. Check every clue against the ticked pairs.

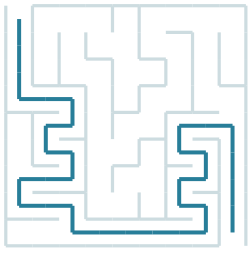


11 • PAGE 13

The winding way

The route is shown in color.

The maze has a single route between its entrance and exit.

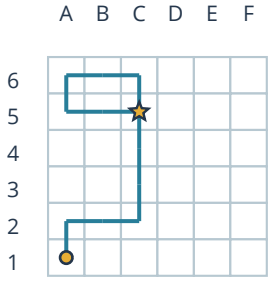


12 • PAGE 14

Coordinates in motion

A1

Read the column letter first and then the row number.



Check your discoveries

Puzzles 13–16 • Use the page number to find each puzzle.

13 • PAGE 15

How many different paths?

3 routes

S-A-D-F-T; S-B-C-E-T; S-B-C-F-T

14 • PAGE 16

Share the mosaic

10 blank tiles

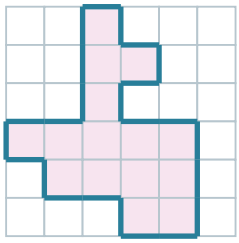
Dotted: 6. Striped: 12. Subtract both from 28. Any nonoverlapping positions work.

15 • PAGE 17

Walk around the edge

24 units

Count each exposed square edge once. Do not count edges between two shaded squares.



16 • PAGE 18

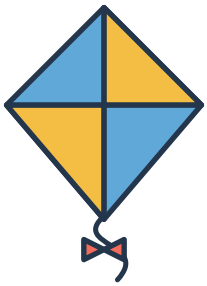
The shifted picture code

STORY

Add 5 to each key number, then use the alphabet strip. Read the five decoded letters in order.

CERTIFICATE OF

Puzzle Power!



Awarded to

_____

for looking closely, thinking carefully,
and trying new ways to solve a puzzle.

Your ideas matter.

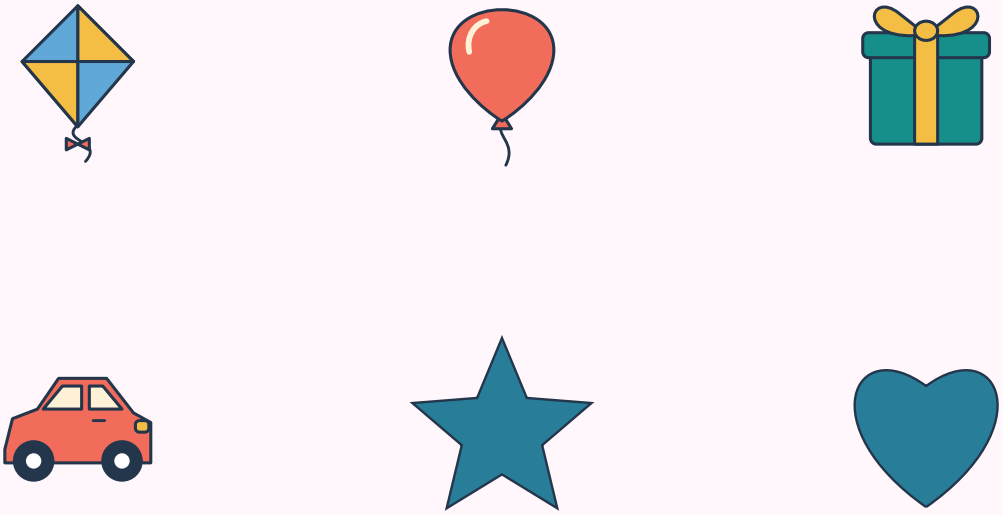
_____ _____

DATE **CHEERED ON BY**

KEEP YOUR CURIOSITY GROWING

Rainbow Carnival

Brain Adventurers



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Come back to a favorite challenge.

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ARLADI PUZZLES